Our assessment of the data supports that the Applicant established the analytical component of the scientific bridge between ABP 654, US-Stelara, and EU-Stelara, using the same methods and statistical approaches used to evaluate that ABP 654 is highly similar to US-Stelara. This supports the relevance of the data generated from clinical studies using EU-Stelara as the comparator for the assessment of biosimilarity.

B. Comparative Analytical Results

Table A. Quality Attributes Analyzed in the Comparative Analytical Assessment

Physico-chemical/Functional Characteristics	Quality Attribute Assessed	Supports a Demonstration of Highly Similar?	Supports the Analytical Component of the Scientific Bridge?
Primary Structure	Amino acid sequence	Yes	Yes
	Intact molecular weight	Yes	Yes
	Reduced and deglycosylated mass of HC and LC	Yes	Yes
Post Translational Modifications	Methionine and tryptophan Oxidation	Yes	Yes
	Deamidation	Yes	Yes
	HC Glycation	Yes	Yes
	Isomerization	Yes	Yes
	N-terminal pyroglutamate	Yes	Yes
	C-terminal Lys truncation	Yes	Yes
Purity and impurities	Size variants by SEC-HPLC	Yes	Yes
	Size variants by reduced CE-SDS	Yes	Yes
	Size variants by non-reduced CE-SDS	Yes	Yes
	Size variants by SE-HPLC-LS and SV-AUC	Yes	Yes
	Non-glycosylated HC	Yes	Yes
	Charge variants by CE-HPLC	Yes	Yes
Glycosylation	Glycan map	Yes	Yes
	Afucosylation	Yes	Yes
	High Mannose	Yes	Yes
	β-Galactosylation	Yes	Yes
	α-Galactosylation	Yes	Yes
	Sialylation: levels and types (NANA, NGNA)	Yes	Yes
	Other glycans G0F, G1F, G2F	Yes	Yes
High Order Structure	Disulfide structure	Yes	Yes

Physico-chemical/Functional Characteristics	Quality Attribute Assessed	Supports a Demonstration of Highly Similar?	Supports the Analytical Component of the Scientific Bridge?
	Free thiols	Yes	Yes
	Secondary structure	Yes	Yes
	Tertiary Structure	Yes	yes
	Thermal Stability	Yes	Yes
Biological activities	Inhibition of IL-23 mediated STAT signaling	Yes	Yes
	Inhibition of IL-12 mediated STAT signaling	Yes	Yes
	Inhibition of binding of IL-23 to receptor IL-12Rβ1	Yes	Yes
	Inhibition of binding of IL-12 to receptor IL-12Rβ1	Yes	Yes
	IL-23 Binding kinetics and affinity	Yes	Yes
	IL-12 Binding kinetics and affinity	Yes	Yes
	-lack of binding to receptor-bound IL-23 and IL-12 on cell surface; -Inhibition of IL-23-induced and IL-12 inducted IFN-ɣ release in PBMCs from a healthy donor; - inhibition of IL-23-induced STAT3 and STAT4 signaling in PBMCs from a healthy donor, binding specificity to cytokines related to IL-12/IL-23,	Yes	Yes
Bioactivity – Fc mediated	Lack of Antibody-dependent cell-mediated cytotoxicity (ADCC)	Yes	Yes
	Lack of Complement-dependent cytotoxicity (CDC)	Yes	Yes
	Lack of ADCP activity	Yes	Yes
	FcRn binding	Yes	Yes
	FcγRIa binding	Yes	Yes
	FcγRIIa (R131) binding	Yes	Yes
	FcγRIIb binding	Yes	Yes
	FcγRIIIa (V158) binding	Yes	Yes